

Multi-Objective Optimization for Inverse Kinematics of Robotic Arm with Joint Configuration

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Abstract—The hardware configuration of modern robots is rapidly developing, yet the fundamental problem of inverse kinematics is still mostly based on analytical or numerical solutions applied to mainstream robotic arms. These traditional methods for inverse kinematics result in an infinite number of solutions when encountering singularities, leading to uncertainty in which solution the robotic arm should use. In this paper, the inverse kinematics of two collaborative robot arm models are investigated using a multi-objective optimization weighted sum method and a particle swarm optimization algorithm. The objective functions include the position and orientation errors of the end-effector and the total rotation amount of the robotic joints. Experimental results show that including a total rotation amount of robotic joints in the objective functions effectively reduces energy consumption. Finally, through simulations and physical model validations according to a series of trajectories obtained, the developed algorithm is proven to achieve the inverse kinematics solution within the allowable error range for engineering requirements.

Keywords—inverse kinematics, robotic arm, multi-objective optimization, particle swarm optimization

I. INTRODUCTION

Industrial robotic arms are increasingly in demand in applications such as mass production and assembly lines, such as car assembly lines, food manufacturing, handling operations, smart logistics chains, and welding operations with the assistance of tools. Among these applications, a trajectory plan is required to guide the robot with needed motions to perform a task, that is, a robot arm needs to achieve a position with respect to a suitable orientation. Such trajectory planning along with the specific orientation of the end effector is required for handling processes in order to grasp objects, such as the welding process and printed circuit board (PCB) soldering process, or for tasks such as peeling bananas and drawing polygons [1]–[3].

Given the position and orientation of the end-effector for the robotic arm, designing the trajectory plan is an inverse problem which can essentially be tackled with the multi-objective optimization [4]–[6].

Traditionally a numerical or analytical method obtains one or more sets of solutions by applying the geometrical parameters and derived formulas of the robotic arm, while further validation is needed for selecting the optimal solution. In many cases, achieving an analytical solution is a tedious and impractical task, especially for a system with a higher degree of freedom [7]–[9]. However, intelligence algorithms, such as particle swarm optimization (PSO) and genetic algorithm (GA), can be applied

to solve inverse kinematics problems with the multi-objective functions to the kinematic problems [10], [11]. Recently, other researchers also proposed the neural-network based solution [12], [13]. These intelligence algorithms and some neural network-based methods for solving kinematic problems are the alternatives to the traditional numerical methods. In practical applications, considering the cost and environmental protection, energy saving can be another goal of inverse kinematics optimization. Therefore, in this study, based on the multi-objective PSO (MOPSO) algorithm [14]–[20], a weighted MOPSO algorithm is developed to include the sum of all the joint rotations of the robotic arm as the third optimization objective in addition to the position and orientation errors of the end-effector, so as to achieve the energy-aware kinematic solution of the robotic arm.

To solve the inverse kinematics of the robotic arm, kinematic modeling has to be carried out first. In this research, we use myCobot 280 arm and UR3 as the platforms [21] and adopt the standard Denavit-Hartenberg (DH) method to establish the coordinate systems for collaborative robots and obtain the DH parameters [22], [23]. The contribution of this paper is listed below:

(1) A weighted multi-objective particle swarm optimization developed including minimizing the sum of rotations for all the robotic joints to solve the inverse kinematics problem; (2) A comparative study on the inverse kinematics of robots, considering the impact of joint rotations on inverse kinematics solutions before and after optimization; (3) The performances of the proposed algorithm for inverse kinematics solutions are compared according to typical trajectories used in the literature [14]; (4) The path plans obtained from the developed algorithms are validated through the simulation environment and the robotic arm (myCobot 280 Pi) for position, orientation, and joint rotations.

II. ROBOTIC ARM MODELING

A basic framework for the robotic arm is to compute the forward kinematics. The forward kinematics is able to calculate the position and orientation of the end-effector based on the coordinate system established between the links. Inverse kinematics is to determine the rotation angles of each joint of the robotic arm to reach a position and end-effector orientation when the position and orientation are specified.

A. Denavit-Hartenberg Parameters

In order to calculate the relative position and orientation between two links, Denavit-Hartenberg (DH) parameters enable

the modeling of the forward kinematics of a robotic arm by establishing a coordinate system for each joint according to certain rules [22], [24]. In this research, since the myCobot 280 Pi and UR3 have similar collaborative robotic arm structures, their coordinate systems can be established as shown in Fig. 1 with the standard DH parameters listed in Table I. For myCobot 280 Pi, the specific parameter values (mm) are $d_1 = 131.56$, $d_3 = 66.39$, $d_4 = 73.18$, $d_5 = 43.6$, $a_2 = 110.4$, and $a_3 = 96$. For UR3, the specific parameter values (mm) are given as $d_1 = 151.19$, $d_3 = 112.35$, $d_4 = 85.35$, $d_5 = 81.9$, $a_2 = 243.65$, and $a_3 = 213.25$.

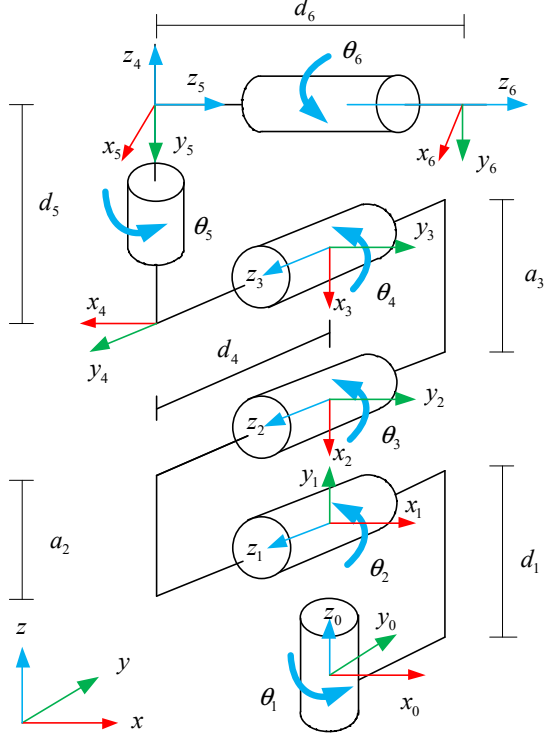


Fig. 1. Cartesian coordinates for DH model

TABLE I. STANDARD DH PARAMETERS FOR MYCOTBOT 280 AND UR3

Joint	θ (rad)	d (mm)	a (mm)	α (rad)
1	θ_1	d_1	0	$\pi/2$
2	$\theta_2 - \pi/2$	0	$-a_2$	0
3	θ_3	0	$-a_3$	0
4	$\theta_4 - \pi/2$	d_4	0	$\pi/2$
5	$\theta_5 + \pi/2$	d_5	0	$-\pi/2$
6	θ_6	d_6	0	0

B. Forward Kinematics

With the DH parameters, the specified position and orientation of the end-effector of a 6-DOF robotic arm, the forward kinematics solution, can be obtained by a series of transformations in (1).

$${}^0T_6 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5 {}^5T_6 = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

where iT_j is the transform [22] from frame i to frame j, which is determined via ith row in Table I. $[p_x \ p_y \ p_z]^T$ is the position vector, and the three unit vectors of $[n_x \ n_y \ n_z]^T$, $[o_x \ o_y \ o_z]^T$, and $[a_x \ a_y \ a_z]^T$ are designated as the normal, orientation, and approach vectors for the orientation of the end effector.

With the establishment of the forward kinematics in (1), the optimal solution is achieved via intelligent optimization algorithms to find the required joint angles ($\theta_1 \sim \theta_6$) based on the forward kinematic model of the robotic arm by minimizing the multi-objective functions.

III. MULTI-OBJECTIVE OPTIMIZATION WITH JOINT CONFIGURATION

A. Multi-Objective Functions

In this work, we adopt the PSO algorithm to minimize three objective functions [25]-[27]. It is common to minimize the position and orientation errors. In order to save the robotic operating energy, the path from each joint rotation can further be considered. Note that the position error, which first objective function, $f_1(\theta_1, \dots, \theta_6)$, is defined as the Euclidian distance between the actual position of the end-effector of the robotic arm and the target location given by (2), where subscript t is used to denote the target. The second objective function, $f_2(\theta_1, \dots, \theta_6)$ for the end-effector orientation error, is defined as the root of the sum of the squared difference of elements between the rotational matrix of the actual orientation of the end-effector and the elements in the target rotational matrix shown in (1) and used in (3). In order to save the robotic arm operating energy, the third objective function is defined as the sum of the rotations of six joints, which is expressed in (4).

$$f_1(\theta_1, \dots, \theta_6) = \sqrt{(x - x_t)^2 + (y - y_t)^2 + (z - z_t)^2} \quad (2)$$

$$f_2(\theta_1, \dots, \theta_6) = \sqrt{\begin{aligned} &(n_x - n_{xt})^2 + (n_y - n_{yt})^2 + (n_z - n_{zt})^2 \\ &+ (o_x - o_{xt})^2 + (o_y - o_{yt})^2 + (o_z - o_{zt})^2 \\ &+ (a_x - a_{xt})^2 + (a_y - a_{yt})^2 + (a_z - a_{zt})^2 \end{aligned}} \quad (3)$$

$$f_3(\theta_1, \dots, \theta_6) = \sum_{i=1}^6 |\theta_i| \quad (4)$$

Note that the second objective function in (3) to minimize the errors of orientation matrix is selected to have flexibility for searching the solution of orientation.

B. Weighted Multi-Objective Particle Swarm Optimization

A weighted summation method is used to combine multi-objective functions and the multi-objective function is optimized via the Particle Swarm Optimization algorithm [28], [29], i.e. weighted multi-objective optimization PSO (WMOPSO). The weighted multi-objective function is expressed as

$$f(\theta_1, \dots, \theta_6) = w_1 f_1(\theta_1, \dots, \theta_6) + w_2 f_2(\theta_1, \dots, \theta_6) + w_3 f_3(\theta_1, \dots, \theta_6) \quad (5)$$

and the following condition is imposed:

$$w_1 + w_2 + w_3 = 1 \quad (6)$$

The modified algorithm is depicted below:

Algorithm WMOPSO for Inverse Kinematics

Set target position P_t and target orientation O_t
Initialize objective function values, weights and solution stores
Initialize PSO parameters
Check joint range limitations
Iterate over weight combinations
for $w_1 = 0.001$ **to** 0.005 **step** 0.0005 **do**
 for $w_3 = 0.001$ **to** 0.005 **step** 0.0005 **do**
 $w_2 = 1 - w_1 - w_3$
 if $w_2 > 0$
 Execute PSO to obtain θ_i ($i = 1, \dots, 6$)
 Calculate position matrix P and orientation matrix O , (P, O)
 Calculate f_1, f_2, f_3
 if f_1 and $f_2 \leq$ allowable limits
 store objective function values, weights and solution
 end if
 end if
 end for
end for

Note: the ranges of w_1 and w_3 are set based on the empirical experience.

IV. EXPERIMENTS AND RESULTS

The experimental section is divided into two portions. The first portion is to solve the inverse kinematics for a specified single position point while the other portion is to solve the inverse kinematics for typical trajectories, that is, the circle, triangle, and sinusoid using the proposed WMOPSO algorithm on both myCobot 280 Pi and UR3 robotic arm structures.

A. Inverse kinematic solution of a single position point

The process of solving for the inverse kinematics of a single position point is to find the rotation angle of each joint of the robotic arm from the initial pose and end-effector orientation to the target position and orientation.

The algorithm first finds a set of solutions, with different weights in the weighted objective function. Fig. 2 shows the dominated particles (dominated solutions) and non-dominated particles (non-dominated solutions) which forms the Pareto optimal front for the WMOPSO algorithm with the myCobot 280 model [30]. According to the Pareto optimal front points, the best solution is the Pareto front point with the minimum weighted objective function (closest to the knee point). The utopia point is the point that corresponds to the minimum of f_1 , f_2 , and f_3 , respectively, which indicates the point of the ideal solution.

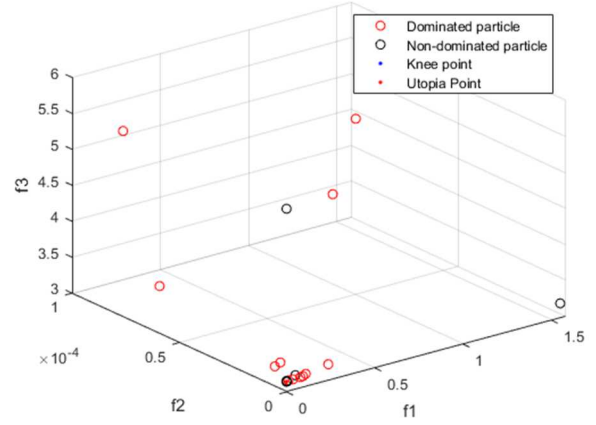


Fig. 2. The best run for Pareto front of WMOPSO myCobot 280 Pi.

Table II shows the comparison results from the multi-objective consisting of weighted sum objective functions $f_1(\theta_1, \dots, \theta_6)$ and $f_2(\theta_1, \dots, \theta_6)$ only, and the proposed weighted sum multi-objective function in (5).

TABLE II. PERFORMACE COMPARISON BEFORE AND AFTER ADDING TO THE WMOPSO ALGORITHM (AVERAGE FOR 10 RUNS)

Algorithm	Position error (m)	Orientation error	Joint rotation mean (rad)
W2OPSO-mycobot280 (f_1+f_2)	0	1.3687×10^{-16}	5.5489
W2OPSO-UR3 (f_1+f_2)	0	3.1265×10^{-16}	4.8687
WMOPSO-mycobot280	0	1.5564×10^{-16}	3.1416
WMOPSO-UR3	0	3.1265×10^{-16}	4.8687

From Table II, it can be seen that our proposed algorithm demonstrates the reduced requirement for the total amount of joint rotations.

B. Inverse kinematics solutions for trajectories

Three trajectories of circle, circle with end-effect with orientation of 45 degrees, and sinusoid are adopted to validate the performance of the algorithms. Table III lists the spatial coordinates and sizes of the trajectories. The sinusoidal trajectory is created using (7). Note that this paper is focused on validation for the algorithm solution and the time consideration for computation for the inverse kinematics algorithm will be considered in our future work.

TABLE III. TRAJECTORY INFORMATION

Trajectory	Center/Start point (x, y, z) for myCobot 280 Pi (mm)	Center/Start point (x, y, z) for UR3 (mm)	Size	Number of Points
Circle	(100, 100, 200)	(200, 200, 200)	$r=50$ (mm)	$n=50$
Circle/	(150, 150, 200)	(200, 200, 200)	$r=50$ (mm)	$n=50$
Sine	(100, 100, 200)	(200, 200, 200)	$A=100$ (mm) $f = 1/\pi$ $\phi = 2$ $T=100$	$n=50$

^a NOTE: The “/” indicates the end effector with 45 degrees downward.

$$\begin{aligned}
x_i &= iT/n + C(1), i = 1, 2, \dots, n \\
y_i &= A \sin(2\pi f_i T/n + \phi) + C(2) \\
z_i &= A \cos(2\pi f_i T/n + \phi) + C(2) - T
\end{aligned} \tag{7}$$

Figs. 3 and 4 plot the performance of the cycle trajectory and the paths of all the joint angles from the WMOPSO for myCobot 280 Pi with the default orientation, that is, vertical downward orientation and Figs. 5 and 6 display the performance results of the end-effector with myCobot 280 pi model moving along the same circular trajectory but with a 45-degree downward orientation instead.

Finally, Figs. 7 and 8 depict the results of the end-effector of the myCobot 280 Pi model operating along a sinusoidal trajectory with a vertical downward orientation.

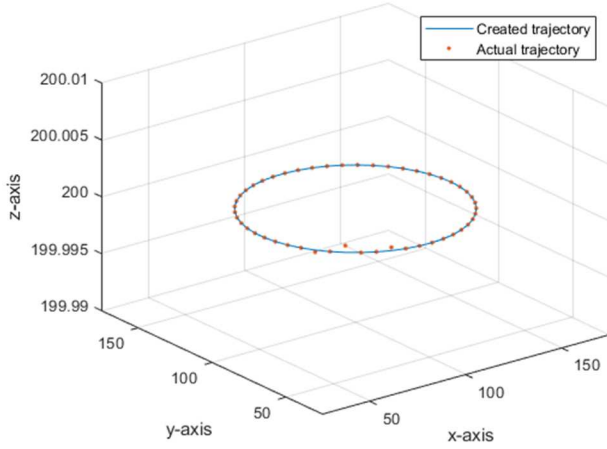


Fig. 3. Circle trajectory performance of the WMOPSO-myCobot 280 Pi.

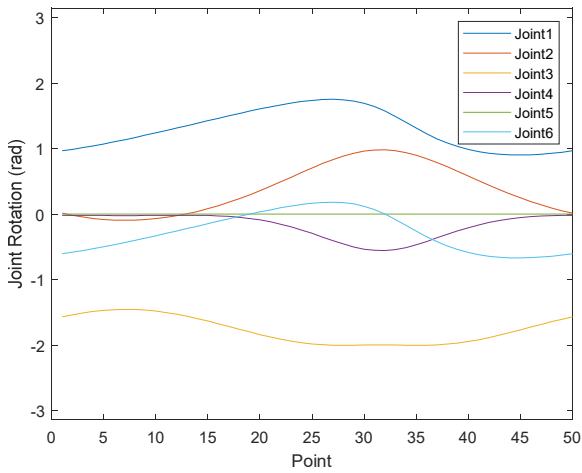


Fig. 4. Circle Paths for six joint angles for the WMOPSO-myCobot 280 Pi.

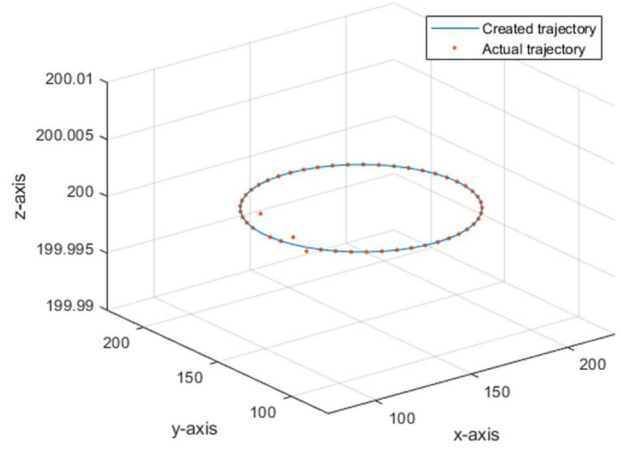


Fig. 5. Circle trajectory performance (45 degree downward) of the WMOPSO-myCobot 280 Pi.

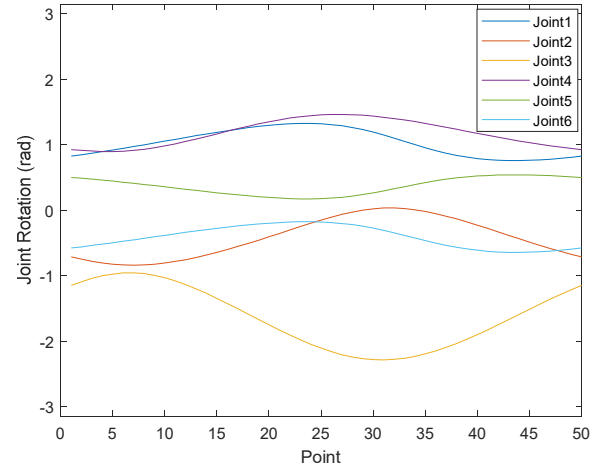


Fig. 6. Circle paths (45 degree downward) for six joint angles for the WMOPSO-myCobot 280 Pi.

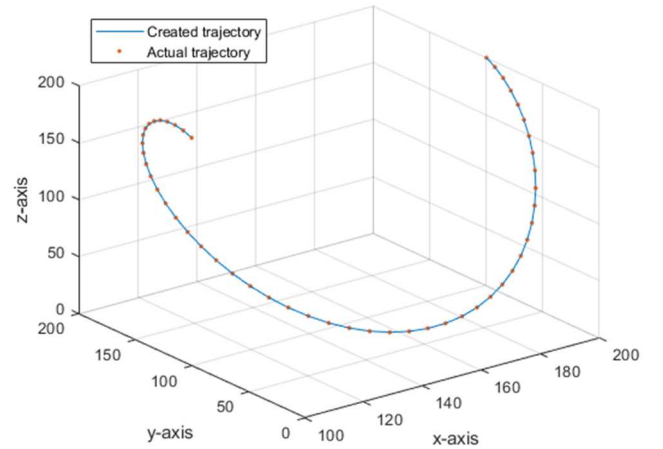


Fig. 7. Sine trajectory performance of the WMOPSO-myCobot 280 Pi.

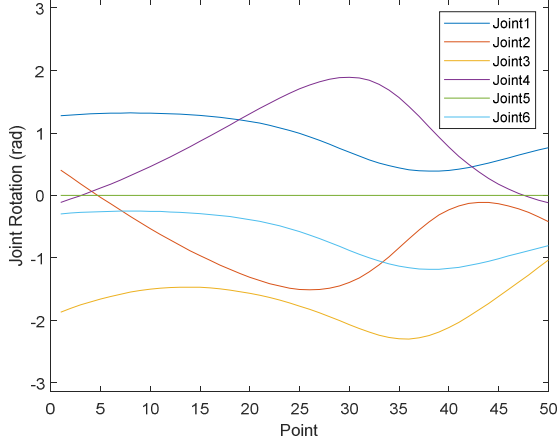


Fig. 8. Sine paths for six joint angles for the WMOPSO-myCobot 280 Pi.

Table IV shows the performance comparisons of the WMOPSO algorithm for three trajectories. Note that the total amount of joint rotations in Table IV is the average value over the total number of points on the trajectory.

TABLE IV. PERFORMANCE COMPARISONS OF WMOPSO

Trajectory	Algorithm-Model	RSE of Position (m)	RSE of Orientation	Total Joint Rotations (rad)
Circle	WMOPSO-myCobot280	7.6068×10^{-8}	4.9398×10^{-8}	3.9735
	WMOPSO-UR3	5.1787×10^{-18}	1.9430×10^{-16}	4.3980
Circle/	WMOPSO-myCobot280	6.9991×10^{-8}	1.6162×10^{-8}	5.0374
	WMOPSO-UR3	5.1159×10^{-18}	2.3007×10^{-16}	6.1968
Sine	WMOPSO-myCobot280	1.7547×10^{-13}	1.1854×10^{-13}	4.9485
	WMOPSO-UR3	4.7474×10^{-18}	2.0211×10^{-16}	5.1709

Table V. lists the comparison between the WMOPSO algorithm developed and the multi-objective modified PSO (MO-M-PSO) algorithm from reference [14] on circular trajectories with the same number of points.

TABLE V. PERFORMANCE COMPARISON BETWEEN THE DEVELOPED ALGORITHM AND EXISTING ALGORITHM IN THE REFERENCE [14]

Algorithm	Position error (m)	Orientation error
MO-M-PSO (Best)	2.31×10^{-6}	1.85×10^{-8}
WMOPSO (Mean)	3.8034×10^{-8}	2.4699×10^{-8}

In comparison, the developed WMOPSO algorithm tested on different robotic arm models has a smaller mean position

error than the best result of the MO-M-PSO algorithm. For the orientation error, the two algorithms are not compared because they use different rotation representations, but they are basically at the same error level.

Verification via simulations in the MATLAB environment was conducted for myCobot 280 and UR3 models [31], respectively. A typical scenario for the myCobot 280 model with the circle trajectory, which has the orientation of 45 degrees downward, is depicted in Fig. 9.

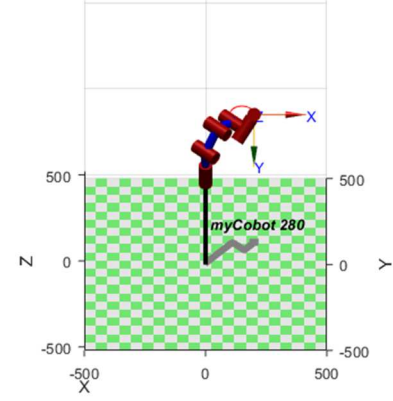
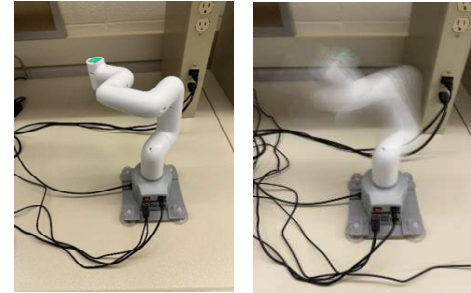


Fig. 9. Trajectory simulation verification on myCobot 280 Pi for circle trajectory with 45 degrees downward.

Finally, the trajectories obtained by the algorithm were all verified using the actual robotic arm. The three trajectories respectively correspond to the actual trajectory results shown in Figs 3, 5, and 7. A typical scenario for the trajectory of circle with 45 degree downward is shown in Fig. 10.



(a) Initial state (b) Operation state

Fig. 10. Physical robotic arm validation for circle trajectory with 45 degrees downward.

V. CONCLUSION

In this paper, a multi-objective function is introduced to solve inverse kinematics problems based on the PSO algorithm using a weighted sum method. The objective functions include the end-effector position errors and orientation errors, and the sum of joint rotations to minimize robotic operating energy. The trajectories solved by the proposed algorithms are verified by simulations and validated by the physical robotic arm. Although the processing using the WMOPSO requires very high computing power, the achieved inverse kinematic solutions

show promising performances and have a potential for applications.

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